Recitation 4

Section: A, B

Problem 1.

The CDF of random variable W is

$$F_w(w) = \begin{cases} 0 & w < -5 \\ \frac{w+5}{8} & -5 \leq w < -3 \\ \frac{1}{4} & -3 \leq w < 3 \\ \frac{1}{4} + \frac{3(w-3)}{8} & 3 \leq w < 5 \\ 1 & w \geq 5. \end{cases}$$

a. What is $P[W \leq 4]$? $P(W \leq 4) = F_w(4) = \frac{1}{4} + \frac{3}{8} = \frac{5}{8}$

b. Find $P[-2 < W \leq 2]$.? $= F(2) - F(-2) = \frac{1}{4} - \frac{1}{4} = 0$

c. What is $P[W > 0]$? $= 1 - P(W \leq 0) = 1 - F(0) = \frac{3}{4}$

d. What is the value of a such that $P[W \leq a] = 1/2$? (a=11/3)

Solution 1.

a. $P[W \leq 4] = F_W(4) = 1/4 + 3/8 = 5/8$

b. $P[-2 < W \leq 2] = F_W(2) - F_W(-2) = 1/4 - 1/4 = 0$

c. $P[W > 0] = 1 - P[W \leq 0] = 1 - F_W(0) = 3/4$

d. By inspection of $F_W(w)$, we observe that $P[W \leq a] = F_W(a) = 1/2$ for this range $3 \leq a < 5$.

In this range,

$F_W(a) = 1/4 + 3(a-3)/8 = 1/2.$

This implies a = 11/3.

Problem 2.

The cumulative distribution function of random variable X is

$$F_x(x) = \begin{cases} 0 & x < -1, \\ (x+1)/2 & -1 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

Find the PDF $f_x(x)$ of X.

Solution 2.

From the CDF, we can find the PDF by direct differentiation. The CDF and corresponding PDF are

$$F_x(x) = \begin{cases} 0 & x < -1, \\ (x+1)/2 & -1 \leq x \leq 1, \\ 1 & x > 1. \end{cases}$$

$$f_x(x) = \begin{cases} 1/2 & -1 \leq w < 1, \\ 0 & otherwise. \end{cases}$$

Problem 3.

Random variable X has PDF

$$f_x(x) = \begin{cases} 1/4 & -1 \leq w \leq 3, \\ 0 & otherwise. \end{cases}$$

Define the random variable Y by Y $= h(X) = X^2$

a. Find E[X] and Var[X]. $E(X) = 1$ $Var(X) = E(X^2) - E^2(X) = \frac{1}{4} + \frac{1}{4} + 1 + \frac{9}{4} = \frac{15}{4} - 1 = \frac{11}{4}$

b. Find h(E[x]) and E[h(X)]. $\rightarrow E(X^2) = \frac{15}{4}$ $\rightarrow h(E(X)) = 1$

c. Find E[Y] and Var[Y].

Solution 3.

$$f_x(x) = \begin{cases} 1/4 & -1 \leq w \leq 3, \\ 0 & otherwise. \end{cases}$$

We recognize that X is a uniform random variable from [-1,3].

a. E[X] = 1 and Var[X] = $\frac{(3+1)^2}{12} = \frac{4}{3}$

b. The new random variable Y is defined as $Y = h(X) = X^2$. Therefore $h(E[X]) = h(1) = 1$ and
$E[h(X)] = E[X^2] = Var[X] + E[X^2] = 4/3 + 1 = 7/3$.

c. Finally
$E[Y] = E[h(X)] = E[X^2] = 7/3$,
$Var[Y] = E[X^4] - E[X^2]^2 = \int_{-1}^{3} \frac{x^4}{4} dx - \frac{49}{9} = \frac{61}{5} - \frac{49}{9}$

Problem 4.

The probability density function of random variable Y is

$$f_Y(y) = \begin{cases} y/2 & 0 \leq y < 2, \\ 0 & otherwise. \end{cases}$$

What are E[Y] and Var[Y]?

Solution 4.

We can find the expected value of X by direct integration of the given PDF.

$$f_Y(y) = \begin{cases} y/2 & 0 \leq y < 2, \\ 0 & otherwise. \end{cases}$$

The expectation is

$E[Y] = \int_0^2 \frac{y^2}{2} dy = 4/3$

To find the variance, we first find the second moment

$E[Y^2] = \int_0^2 \frac{y^3}{2} dy = 2$

The variance is then $Var[Y] = E[Y^2] - E[Y]^2 = 2 - (4/3)^2 = 2/9$.

Problem 5.

The cumulative distribution function of random variable Y is

$$f_y(y) = \begin{cases} 0 & y < -1, \\ (y+1)/2 & -1 \leq y \leq 1, \\ 1 & y > 1. \end{cases}$$

What are E[Y] and Var[Y]?

Solution 5.

a. We can find the expected value of Y by first differentiating the above CDF to find the PDF

$$f_y(y) = \begin{cases} 1/2 & -1 \leq y \leq -1, \\ 0 & otherwise. \end{cases}$$

It follows that

$E[Y] = \int_{-1}^1 \frac{y}{2} dy = 0.$

b. $E[Y^2] = \int_{-1}^1 \frac{y^2}{2} dy = 1/3,$

$Var[Y] = E[Y^2] - E[Y]^2 = 1/3 - 0 = 1/3.$

Problem 6.

The cumulative distribution function of random variable U is

$$F_U(u) = \begin{cases} 0 & u < -5, \\ \frac{u+5}{8} & -5 \leq u < -3, \\ \frac{1}{4} & -3 \leq u < 3, \\ \frac{3u-7}{8} & 3 \leq u < 5, \\ 1 & u \geq 5. \end{cases}$$

a. What are E[U] and Var[U]?

b. What is $E[2^U]$?

Solution 6.

PDF is

$$F_U(u) = \begin{cases} 0 & u < -5, \\ \frac{1}{8} & -5 \leq u < -3, \\ 0 & -3 \leq u < 3, \\ \frac{3}{8} & 3 \leq u < 5, \\ 0 & u \geq 5. \end{cases}$$

a. The expected value of U is

$E[U] = \int_{-\infty}^{\infty} u f_U(u) du = \int_{-5}^{-3} \frac{u}{8} du + \int_{3}^{5} \frac{3u}{8} du = 2$

The second moment of U is

$E[U^2] = \int_{-\infty}^{\infty} u^2 f_U(u) du = \int_{-5}^{-3} \frac{u^2}{8} du + \int_{3}^{5} \frac{3u^2}{8} du = 49/3$

The variance of U is $Var[U] = E[U^2] - (E[U])^2 = 37/3$.

b. Note that $2^U = e^{(ln2)U}$. This implies that

$\int 2^u du = \int e^{(ln2)u} du = \frac{1}{ln2} e^{(ln2)u} = \frac{2^u}{ln2}$

The expected value of 2^U is then

$E[2^U] = \int_{-\infty}^{\infty} 2^u f_U(u) du = \int_{-5}^{-3} \frac{2^u}{8} du + \int_{3}^{5} \frac{3*2^u}{8} du = \frac{2^u}{8ln2}|_{-5}^{-3} + \frac{3*2^u}{8ln2}|_{3}^{5} = \frac{2307}{256ln2} = 13.001$

Problem 7.

Three people toss a fair coin and the odd one pays for coffee. If the coins all turn up the same, they are tossed again. Find the probability that fewer than 3 tosses are needed.

Solution 7.

The probability that all coins turn up the same is $2/8 = 0.25$. Therefore, for using geometric distribution, $p = 0.75$ and $q = 0.25$, and hence we have $P(X < 3) = P(X = 1) + P(X = 2) = 0.75 + 0.1875 = 0.9375$.

Problem 8.

The probability that a person will die when he or she contracts a viral infection is 0.004. Of the next 4000 people infected, what is the mean number who will die?

Solution 8.

$\mu = (4000)(0.004) = 16$.

Problem 9.

The acceptance scheme for purchasing lots containing a large number of batteries is to test no more than 75 randomly selected batteries and to reject a lot if a single battery fails. Suppose the probability of a failure is 0.001.

(a) What is the probability that a lot is accepted?

(b) What is the probability that a lot is rejected on the 20th test?

(c) What is the probability that it is rejected in 10 or fewer trials?

Solution 9.

$n = 75$ with $p = 0.999$.

(a) $X =$ the number of trials, and $P(X = 75) = (0.999)^{75}(0.001)^0 = 0.9277$.

(b) $Y =$ the number of trials before the first failure (geometric distribution), and $P(Y = 20) = (0.001)(0.999)^{19} = 0.000981$.

(c) $1 - P(\text{no failures}) = 1 - (0.001)^0(0.999)^{10} = 0.01$.